

Project Id	AES/KVGH/STP/2026-27/111
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Sheet	1 of 1

CONTROL PANEL LOAD LIST

1.5 MLD Sewage Treatment Plant • KVG Medical College & Hospita, Sullia

Client / Project	KVG Medical College & Hospital	Panel	STP Main Control Panel (MCC cum PLC)
System	1.5 MLD Sewage Treatment Plant (STP)	Supply	415 V ±10%, 3Ph, 4W, 50 Hz
Location	Sullia	Control Voltage	230 V AC / 24 V DC

CONNECTED LOAD (HP)	CONNECTED LOAD (kW)	OPERATING LOAD (HP)	OPERATING LOAD (kW)
298.75	222.79	171.25	127.71

A. POWER OUTPUTS – MOTOR & ELECTRICAL LOADS

Sl. No.	Equipment Description	Rating (HP)	Rating (kW)	Supply	Qty (Nos.)	Duty	Connected Load (HP)	Connected Load (kW)	Operating Load (kW)
1	Feed Pump (Working) – Submersible	7.50	5.59	415V, 3Ph	1	Working	7.50	5.59	5.59
2	Feed Pump (Standby) – Submersible	7.50	5.59	415V, 3Ph	1	Standby	7.50	5.59	–
3	Permeate Pump 1 (Working)	5.00	3.73	415V, 3Ph	1	Working	5.00	3.73	3.73
4	Permeate Pump 1 (Standby)	5.00	3.73	415V, 3Ph	1	Standby	5.00	3.73	–
5	Motorized Valve – MV1	0.25	0.19	230V, 1Ph	1	Working	0.25	0.19	0.19
6	Permeate Pump 2 (Working)	5.00	3.73	415V, 3Ph	1	Working	5.00	3.73	3.73
7	Permeate Pump 2 (Standby)	5.00	3.73	415V, 3Ph	1	Standby	5.00	3.73	–
8	Motorized Valve – MV2	0.25	0.19	230V, 1Ph	1	Working	0.25	0.19	0.19
9	Backwash Pump	5.00	3.73	415V, 3Ph	1	Working	5.00	3.73	3.73
10	Motorized Valve – MV3	0.25	0.19	230V, 1Ph	1	Working	0.25	0.19	0.19
11	Air Blower – Aeration (Working)	60.00	44.74	415V, 3Ph	1	Working	60.00	44.74	44.74
12	Air Blower – Aeration (Standby)	60.00	44.74	415V, 3Ph	1	Standby	60.00	44.74	–
13	Air Blower – MBR (Working)	50.00	37.29	415V, 3Ph	1	Working	50.00	37.29	37.29
14	Air Blower – MBR (Standby)	50.00	37.29	415V, 3Ph	1	Standby	50.00	37.29	–
15	Dosing Pump	0.50	0.37	230V, 1Ph	1	Working	0.50	0.37	0.37
16	RAS Pump (Submersible)	7.50	5.59	415V, 3Ph	1	Working	7.50	5.59	5.59
17	Sludge Pump – Aeration Tank (Submersible)	5.00	3.73	415V, 3Ph	1	Working	5.00	3.73	3.73
18	Treated Water Pump	7.50	5.59	415V, 3Ph	1	Working	7.50	5.59	5.59
19	Extra Pump – 1	3.00	2.24	415V, 3Ph	1	Working	3.00	2.24	2.24
20	Extra Pump – 2	5.00	3.73	415V, 3Ph	1	Working	5.00	3.73	3.73
21	Extra Pump – 3	7.50	5.59	415V, 3Ph	1	Working	7.50	5.59	5.59
22	Mechanical Screen	2.00	1.49	415V, 3Ph	1	Working	2.00	1.49	1.49
TOTAL					22		298.75	222.79	127.71

B. FIELD INSTRUMENTS – ANALOG / COMMUNICATION INPUTS

Sl. No.	Instrument Description	Signal Type	Supply	Qty (Nos.)
1	Vacuum Transmitter	4–20 mA	230V, 1Ph	1
2	Pressure Transmitter	4–20 mA	230V, 1Ph	1
3	pH Meter	RS-485 (Modbus)	230V, 1Ph	1
4	Electromagnetic Flow Meter (EMF) – Inlet	RS-485 (Modbus)	230V, 1Ph	1
5	Electromagnetic Flow Meter (EMF) – Outlet	RS-485 (Modbus)	230V, 1Ph	1

C. DIGITAL INPUTS

Sl. No.	Input Description	Signal Type	Supply	Qty (Nos.)
1	Float Level Switch – LS1 (Feed Pumps)	Potential-free (NO/NC)	24V DC	1
2	Float Level Switch – LS2 (Feed Pumps)	Potential-free (NO/NC)	24V DC	1
3	Float Level Switch – LS3 (Permeate Pumps)	Potential-free (NO/NC)	24V DC	1
4	Float Level Switch – LS4 (Permeate Pumps)	Potential-free (NO/NC)	24V DC	1
5	Float Level Switch – LS5 (Backwash Pump)	Potential-free (NO/NC)	24V DC	1
6	Float Level Switch – LS6 (TW Pump)	Potential-free (NO/NC)	24V DC	1

D. PANEL ACCESSORIES / ADDITIONAL REQUIREMENTS

Sl. No.	Description	Type	Supply	Qty (Nos.)
1	Panel Illumination Lamp with Door Limit Switch	LED	230V, 1Ph	1
2	Voltmeter with Selector Switch	Digital	415V, 3Ph	1
3	Energy Meter	Digital, RS-485	415V, 3Ph	1
4	3-Pin Switch Socket	6/16 A	230V, 1Ph	2

NOTES

1. Conversion factor: 1 HP = 0.7457 kW.
2. Operating load includes only equipment with Duty = "Working"; Standby loads are excluded.
3. Motor ratings to be verified against final equipment datasheets before panel manufacturing.